

AUTOMATIC HAMNOSYS GENERATION FROM SIGN LANGUAGE VIDEOS USING HANDSHAPE, ORIENTATION, LOCATION, AND MOVEMENT LEARNING WITH AVATAR-BASED VERIFICATION

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1. Problem Statement

To develop an automatic system that generates HamNoSys from Indian Sign Language (ISL) videos by modeling handshape combinations, orientation, location, and movement, enforcing strict Context-Free Grammar (CFG) syntax, and verifying accuracy through 3D avatar-based SiGML rendering.

2. Baseline Work & Approved Mathematical Standards (V1)

- **Custom Dataset & ML Handshape Model:** Built a 149-handshape combination dataset with synthetic augmentation. Used a DAG-based hierarchical model (Basic Handshape, Thumb, Finger Type, Finger Contact) using MediaPipe 21-hand landmark feature vectors.
- **Approved Anatomical Facial Ratios:** Normalized vertical face coordinates ($\text{rel}_y = \frac{y - y_{\text{top}}}{H}$) using standard Da Vinci facial third subdivisions (0.10, 0.18, 0.28, 0.36, 0.44, 0.55, 0.65, 0.777, 0.888) for forehead, eyes, nose, lips, teeth, and chin.
- **Vector Geometry Orientation:** View classification (Signer/Right/Bird) and $360^\circ/8 = 45^\circ$ octant angular partitioning ($\pm 22.5^\circ$ tolerance) mapping 2D/3D wrist-to-MCP vectors into 8 canonical palm directions.
- **Scale-Invariant Body & Arm Ratios:** Normalized torso zones by shoulder width and torso height. Calculated elbow flex angles via vector dot products ($\theta = \arccos\left(\frac{\vec{u} \cdot \vec{v}}{\|\vec{u}\| \|\vec{v}\|}\right)$) and arm extension ratios ($R = \frac{\|S-W\|}{\|S-E\| + \|E-W\|} > 0.95$).
- **Empirical Touch Thresholds:** Distance thresholding $\epsilon = 0.02$ (3.4 cm physical equivalent on 170 cm body scale) to isolate contact versus proximity while filtering MediaPipe landmark jitter.

3. Architectural Limitations & Flaws of Baseline Codebase (V1)

- **Ungrammatical Token Concatenation:** `combine_hamnosys` used naive `set()` deduplication across all frames, producing massive un-structured string blobs (27–33 tokens) containing conflicting orientations, raw finger states, and non-HamNoSys tags (e.g., `signer`).
- **Low Precision & High False Positives:** Unfiltered tag dumping resulted in low token precision ($\sim 11.2\%$) and low F1 score (~ 0.14).
- **SiGML / Avatar Parser Rejection:** Avatar generators (JASigning) rejected V1 outputs due to lack of Context-Free Grammar (CFG) structure.
- **Static Orientation Limitation:** Failed to model dynamic hand rotation transitions during continuous gesture movements (missing `hamreplace` tokens).
- **Environment Lock-in & Import Crashes:** Notebook-centric scripts contained un-guarded `google.colab` dependencies (`cv2.imshow`, Colab paths) causing immediate runtime crashes on local machines.

4. Research Gap & Extended Project Objectives

- **Grammatical Phonetic Compilation:** Build a formal Context-Free Grammar (CFG) parser that enforces strict HamNoSys sequence structure:

$\text{Sign} ::= \text{Handshape} + [\text{ActiveFinger}] + \text{InitialOrientation} + \text{Location} + [\text{Contact}] + [\text{Movement}] + [\text{State}]$

- **Dynamic Temporal State Transitions:** Implement windowed temporal frequency analysis (start vs end frames) to detect rotation and generate `hamreplace` state transitions.
- **Preserve All Standard Math Ratios:** Retain and enforce 100% of the approved anatomical ratios (Da Vinci facial thirds, 45° octants, elbow cosine rule) while enhancing numerical precision.
- **Standalone Execution Engine & Web Stack:** Eliminate Colab dependencies, create a local CLI runner (`run_local.py`), automated SiGML hex converter (`generate_sigml_flow.py`), and a Flask web application.

5. Extended Improvements (V2) – Grammatical CFG Compiler

Baseline (V1) String Concatenation

```
def combine_hamnosys(*modules):  
    final = []  
    for m in modules:  
        final.extend(extract_labels(m))  
    seen = set()  
    clean = []  
    for x in final:  
        if x not in seen:  
            clean.append(x); seen.add(x)  
    return " ".join(clean)
```

Result: 27-33 ungrammatical tokens; invalid syntax.

Extended (V2) CFG Pipeline Compiler

```
# Enforces strict HamNoSys CFG Sequence  
seq = [handshape]  
if active_finger: seq.append(active_finger)  
seq.extend(format_ori(ori_start))  
seq.append(body_location)  
if contact: seq.append(contact)  
if movement: seq.append(movement)  
if ori_start != ori_end:  
    seq.append("hamreplace")  
    seq.extend(format_ori(ori_end))
```

Result: Clean 7-10 token sequence; 100% valid.

6. Extended Improvements (V2) – Dynamic State & 3D Vector Math

- **Dynamic Orientation State-Transition Engine:**

- Divides gesture frames into initial (first 30%) and final (last 30%) temporal windows.
- Computes mode orientations for both windows; appends `hamreplace` + final orientation upon rotation.

- **3D Palm Normal Cross-Product Calibration** (`ori_model2.py`):

- Computes exact 3D palm normal vector straight from MediaPipe landmarks:

$$\vec{n}_{\text{palm}} = (P_{\text{index_mcp}} - P_{\text{wrist}}) \times (P_{\text{pinky_mcp}} - P_{\text{wrist}})$$

- Extended Finger vector: $\vec{v}_{\text{finger}} = P_{\text{middle_tip}} - P_{\text{wrist}}$.
- Resolves camera-relative depth coordinates to accurately distinguish `hampalmd` (palm down/away) vs `hampalmu` (palm up/inward).

7. Extended Improvements (V2) – Standalone Engine & Web Stack

- **Local OS Execution Engine** (`run_local.py`):

- Completely purged all `google.colab` imports, Drive mounts, and notebook code.
- Added standard `argparse` CLI, local path resolution, and C++ log level suppression (`TF_CPP_MIN_LOG_LEVEL=3`). Runs out-of-the-box on Windows/Linux/macOS.

- **Automated SiGML Converter** (`generate_sigml_flow.py`):

- Maps generated HamNoSys tokens to Unicode hex codes via `conversionSpreadSheet.txt`.
- Automatically executes `HamNoSys2SiGML.py` to output valid `.sigml` XML files for 3D avatar rendering.

- **Full-Stack Web Application** (`webapp/app.py`):

- Flask backend and interactive HTML/JS frontends for real-time sign video uploads and visual mapping.

8. Legitimacy: 100% Preservation of Approved Standards

Key Takeaway for Evaluation

We did NOT alter or break any underlying mathematical standards. All approved anatomical ratios, Euclidean spatial formulas, and empirical thresholds established in V1 were strictly preserved and built upon:

Standard Model / Ratio	Mathematical Value	Preservation Status in V2
Facial Rule of Thirds	0.10, 0.18, 0.28, 0.36, 0.44, 0.55, 0.65, 0.777, 0.888	Preserved (Da Vinci lower-third splits)
Orientation Octants	$360^\circ / 8 = 45^\circ (\pm 22.5^\circ \text{ tolerance})$	Preserved (Exact 8-direction partitioning)
Elbow Angle Formula	$\theta = \arccos \left(\frac{\vec{u} \cdot \vec{v}}{\ \vec{u}\ \ \vec{v}\ } \right) < 120^\circ$	Preserved (Euclidean joint dot product)
Arm Extension Ratio	$R = \frac{\ S-W\ }{\ S-E\ + \ E-W\ } > 0.95$	Preserved (Triangle inequality ratio)
Touch Epsilon Threshold	$\epsilon = 0.02 (\approx 3.4 \text{ cm on } 170 \text{ cm scale})$	Preserved (Filters MediaPipe landmark jitter)

9. Quantitative Evaluation & Accuracy Improvements

- **Token Precision Boost (+70% Increase):**

- **Baseline (V1) Precision:** $\sim 11.2\%$ — Severely degraded by hyper-bloated over-prediction (generating 27–34 candidate tokens per 7-token sign).
- **Extended (V2) Precision:** $75.0\% - 88.5\%$ — CFG parser eliminates false positive token clutter, yielding an average **+70% Precision boost**.

- **F1 Score Increase ($> 5\times$ Growth):**

- Increased from ~ 0.14 in V1 to $0.78 - 0.85$ in V2 due to precise token selection.

- **3D Avatar Verification Accuracy (Cosine Similarity):**

- Evaluates physical landmark vector A (avatar rendering) vs B (input signer video):

$$\text{cos_sim}(A, B) = \frac{A \cdot B}{\|A\| \|B\|} = 0.8812 \quad (88.12\% \text{ Gesture Similarity})$$

- Low spatial Root Mean Square Deviation ($\text{RMSD} = 0.2781$).

10. Comparative Execution Results & Accuracy Summary (V1 vs V2)

Metric / Test Case	Baseline (Seniors' V1)	Extended System (Our V2)
Prompt_1.mp4	hamflathand signer hamextfingerr hampalmu hamextfingeru... (27 tokens)	hamflathand hammiddlefinger hamextfingerr hampalmu... (10 CFG tokens)
hello_hi.mp4	hamfist hamflathand hamthumboutmod signer hamextfingerd... (33 tokens)	hamflathand hammiddlefinger hamextfingeru hampalmd... (7 CFG tokens)
Precision (Prec)	~ 11.2% (High false positives)	80.0% – 85.7% (+70% boost)
F1 Score	~ 0.14	0.82 – 0.85 (> 5× increase)
Execution	Crashes locally (google.colab missing)	Runs locally via run_local.py (13.3s – 30.5s)
SiGML / Avatar	Rejected by 3D Avatar Parser	Renders on JASigning (88.12% CosSim)

References I



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Thank You